



Driving Smart Energy Management and Environmental Monitoring Solution for a Leading Global Engineering Services & Solutions Company

Introduction

Large facilities face immense pressure to reduce operational costs, ensure regulatory compliance, ensure continuity of functions, and achieve ambitious sustainability & environmental goals.

This case study highlights how PowerOptimus partnered with a leading global engineering services & solutions company to address these challenges by implementing a comprehensive energy and environment monitoring solution.

It serves as a reference for organizations looking to use smart technologies to gain visibility, improve operational efficiency, and advance their sustainability journey.



Client Overview

Our client is a leading global engineering services and solutions company with over 1,500 employees worldwide. To support their growth and foster innovation, they recently set up a new facility in Hinjewadi, Pune, India, consisting of three distinct buildings:

- Administration Building
- Operations Building (the primary hub)
- Utility Building

The operations building stands out as the centerpiece, designed to accommodate 1,000 employees across 10 floors, each floor tailored for diverse functions such as:

- Research & Innovation Center
- IT Department
- Engineering Services Division
- Software Engineering Services
- Conference Rooms and Cafeteria
- Enclosed parking equipped with EV charging stations

Challenges

As the client's facility scaled up, they faced complex energy & environmental management challenges, including:

- **Monitoring & Managing Energy Consumption Against Committed Demand:**

Difficulty in tracking real-time energy usage and ensuring it stayed within contract demand limits.

- **Finding Consumption Patterns:**

Needed visibility into energy & water consumption across different floors and departments, and understanding how energy was distributed.

- **Improving Power Source Utilization:**

Deciding when to switch between Solar, DG, & Grid power for maximum efficiency and cost savings.

- **Ensuring an Optimum Comfortable Environment:**

Keeping consistent indoor temperature, humidity, and air quality despite varying departmental needs.

- **Maintenance of Critical Equipment:**

Including DG sets, HVAC systems, STP, pumps, and elevators, and ensuring minimal downtime through initiative-taking strategies.

- **Monitoring Air Quality Parameters:**

Such as air circulation efficiency, CO & CO₂ levels, and overall indoor air quality to meet environmental and safety standards.

PowerOptimus Implementation Process

Implementing PowerOptimus, an Energy Management & Environmental Monitoring (EEM) system, has been deployed to provide end-to-end visibility into

- Energy consumption
- Environmental quality and
- Operational efficiency.

The implementation includes a wide network of smart sensors, gateways, and nodes, working over an independent data fabric, securely separated from the facility's IT network.

1. Assessment and Planning

- PowerOptimus team conducted an in-depth audit of building energy needs and asset equipment specification.
- Defined the best locations/equipment for sensor & energy meters placement across floors, focusing on high-consumption and high-risk areas.
- With an as-built BIM (Building Information Modelling) model in place, the assessment phase of implementing a system like PowerOptimus becomes significantly more efficient and correct. It enables precise visualization of the building layout, easy identification of HVAC and MEP systems, detection of high-consumption zones, optimized sensor placement planning, and detailed asset mapping for predictive maintenance strategies



2. Installation and Integration

The following sensors were installed across the facility for multi-parameter monitoring:

- Air Quality Sensors (Qty - 20) – Covering all floors to track particulate matter, VOCs, & IAQ.
- Humidity Sensors (Qty - 40) – Installed in 40 rooms for environmental balance monitoring.
- Temperature & Humidity Sensors (Qty – 80) – Distributed across equipment rooms for thermal profiling.

- Noise/Vibration Sensors (Qty - 14) – Installed on pumps, motors, and DG sets to track predictive maintenance parameters.
- Sixteen Energy Meters were installed at each panel level to capture granular energy consumption.
- Three pH Meters – For monitoring water quality and sewage treatment.

- Two TDS Sensors were installed in STP to measure dissolved solids in water.
- Nearly 22 units of CO₂ & CO Sensors were deployed in parking and work areas for air safety monitoring.
- 34 Lux / Illuminance Sensors were installed in basements and staircases to monitor lighting efficiency in low-light areas.

- Deployed 23 smart IOT nodes and 14 gateways connected to various sensors and meters
- Gateways aggregate data securely, while Nodes extend coverage to prevent blind spots.
- The network operates independently from the facility's IT infrastructure for secure and uninterrupted data capture.

- Tracks energy generation and consumption from solar panels and conventional sources, down to individual equipment like pumps, motors, and HVAC units.
- Tracks renewable vs. non-renewable consumption with precision.
- Integrates with solar panels to capture renewable energy data.

- Enables predictive maintenance, anomaly detection, and energy optimization aligned with sustainability goals.
- All data streams into the PowerOptimus platform for real-time visualization and advanced analytics.
- Supports progress toward sustainability and carbon-reduction goals.

3. Dashboard Configuration and Alerts

- Customized dashboards were set up to help facility managers easily view live consumption, trends, and heat maps insights.
- Set threshold-based alerts to notify operations teams of deviations.

4. Maintenance Strategy Activation

- Configured preventive maintenance schedules based on runtime data.
- Enabled predictive alerts using consumption and wear patterns.

5. Training, Go-Live & Scale

- Conducted workshops for facility and operations teams on system usage.
- Established SOPs for responding to alerts and reviewing dashboards.

6. Monitor, Manage, Scale & Elevate

- Refine objectives and scope for better coverage to reduce costs incrementally.

Solution Overview

The engineering services company selected PowerOptimus to deploy a data-driven energy management system and environmental monitoring system using smart sensors, nodes, and a centralized analytics platform.

Key Capabilities

Digital Twin:

Use Digital Twin technology to create a real-time virtual model of the facility's energy systems, enabling visibility into power flow and equipment performance. This helps identify inefficiencies, optimize loads, simulate improvements, and support smarter, more efficient decision-making.

• Sensor Deployment on Key Assets:

Smart IoT nodes were connected to sensors and meters installed on pumps, motors, elevators, HVAC systems, lighting circuits, and water lines to capture real-time operational data.

• Environmental & Energy Parameters Monitored:

The system tracks energy usage, temperature, humidity, air quality, hazardous gases, and water consumption to build a complete environmental & energy profile of the facility.

• Dynamic Energy Management:

Real-time insights guide when to switch between solar, DG, and grid power for the best cost, efficiency, and sustainability outcome.

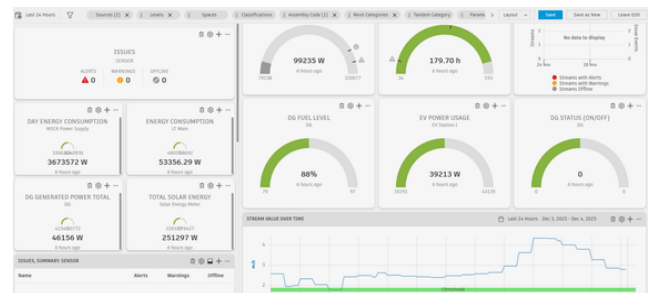


• Data-Driven Operational Optimization:

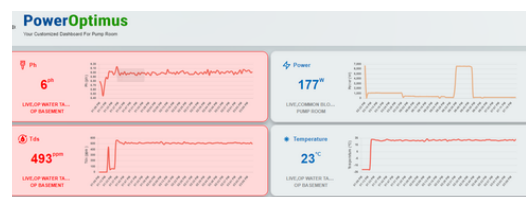
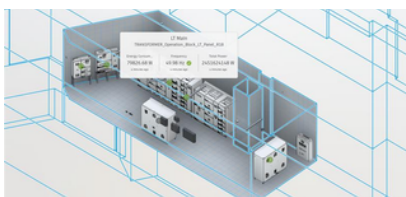
Floor-wise and equipment-level insights enable intelligent alerts, automated thresholds, and recommended adjustments—improving efficiency and reducing operational waste.

• Asset Health (Managing Equipment) & Predictive Maintenance:

Provided insights on when to switch between solar, DG, and grid power, based on demand, cost, and environmental factors.



Results & Benefits Achieved



• Reduction in Energy Consumption by 8%

Optimized HVAC, lighting, and equipment schedules, alongside smarter source switching, directly contributed to lowering energy costs.

- **Reduction in Water Usage**

Identified patterns of excess water use and implemented targeted conservation measures.

- **Seasonal Comfort Optimization**

Automated HVAC adjustments during summer and winter ensured the safer environment with reduced manual interventions.

- **Enhanced Operational Efficiency**

Facility managers now have granular insights into consumption and can make informed decisions, improving overall operational agility.

- **Predictive Maintenance Enabled**

Reduced unexpected downtimes and helped prolong asset life by using data for initiative-taking maintenance.

- **Improved Asset and Facility Management**

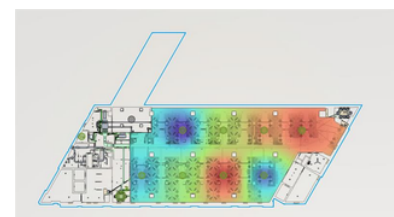
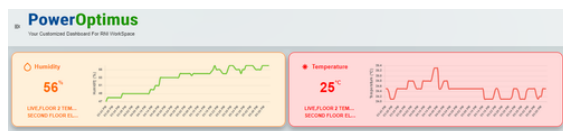
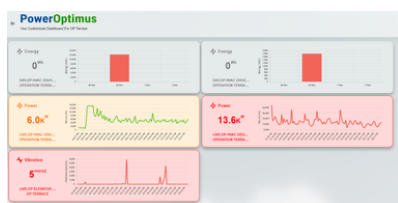
Streamline the management and maintenance of building assets.

- **Lifecycle Management:**

Track and optimize the performance, maintenance, and efficiency of critical systems throughout their operational lifespan to reduce costs, extend asset life, and ensure energy efficiency.

- **Space Utilization:**

Analyze occupancy and usage patterns across facilities to improve space allocation, reduce energy waste, and support better planning for comfort and productivity.



Strategic Takeaways

- **Real-Time Visibility:**

The ability to drill down into Energy distribution from the facility to a granular level of equipment helped the organization to find inefficiencies, check consumption patterns, and take targeted actions for optimization.

- **Data-Driven Sustainability:**

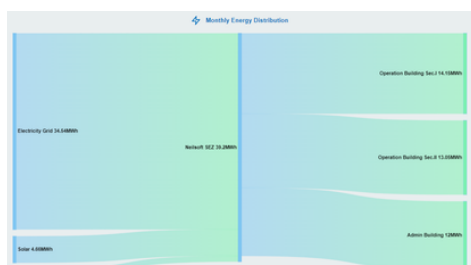
The ability to dynamically balance solar, DG, and grid power reduces carbon footprint and reliance on non-renewable sources.

- **Workplace Well-Being:**

Monitoring of temperature, humidity, air circulation, and hazardous gases ensure compliance with safety norms and improved employee comfort.

- **Scalable Approach:**

As the client continues to expand and add more facilities and locations, the scalability of PowerOptimus will quickly integrate these into its EEMS (Energy & Environment Management System) portfolio. Additionally, this implementation can easily be extended to other facilities such as commercial offices, hospitals, and industrial plants and factories.



How Asset Maintenance Is Critical for Achieving Uninterrupted Operation

Asset maintenance plays a vital role in ensuring continuous, efficient, and safe operations in any facility. Here's why it's critical:

- **Prevents Unexpected Downtime:**

Regular maintenance helps find wear and tear early, reducing the risk of sudden breakdowns that can halt operations.

- **Extends Equipment Life:**

Well-maintained assets perform better for longer, reducing the need for frequent replacements and lowering capital expenditure.

- **Improve Safety & Compliance:**

Initiative-taking maintenance ensures systems run within safety parameters and follow regulatory standards, minimizing operational risks.

- **Enhance Productivity:**

With fewer breakdowns & interruptions, teams can support a steady workflow, improving overall productivity.

- **Reduces Cost of Emergency Repairs:**

Planned maintenance is more cost-effective than reactive fixes and helps refine maintenance budgets.

- **Enables Predictive Maintenance:**

With the use of monitoring tools and analytics, teams can shift from time-based to condition-based maintenance, allowing them to act before failures occur.

In short, asset maintenance is not just a support function; it's a strategic enabler of uninterrupted, high-performance operations.

Learning Objectives

- Understand how integrating IoT sensors with a centralized platform can manage energy consumption and environmental health monitoring with data-driven precision.
- Recognize the importance of real-time monitoring for improving energy efficiency and occupant comfort.
- Explore preventive and predictive maintenance strategies enabled by IoT data.
- Evaluate how multi-source power systems (solar, DG, grid) can be managed for cost and sustainability goals.
- Learn a scalable implementation roadmap that can be adapted across diverse types of buildings.

Summary

This implementation by Power Optimus has transformed how a leading engineering services company manages its energy, environmental, and maintenance operations. By integrating smart IoT nodes and advanced sensor frameworks, the client achieved:

- Significant cost savings
- Better sustainability performance
- Improved employee comfort and safety
- Enhanced lifecycle management of critical assets

The success of this project highlights how commercial buildings, whether offices, residential complexes, or hospitals, can use similar smart monitoring solutions to achieve their operational and sustainability goals.



Neil Automation Pvt. Ltd.

Pride Parmar Galaxy, 8th Floor, 10/10 + A, Sadhu Vaswani Chowk, Pune 411001, India
Inquiry: eems@neilautomation.com | Tel: +91 86000 26806

India Sales Offices : Bangalore | New Delhi | Mumbai | Ahmedabad



<https://poweroptimus.com/>



eems@neilautomation.com